

Section 2.5

71.

- a. We want to obtain $P(B'|A')$. since A and B are independent, A' and B' are also independent.

$$\text{Thus } P(B'|A') = P(B') = 1 - P(B) = 0.3$$

- b. Let C denote the event that at least one of two projects will be successful. Then $C = A \cup B$

$$P(C) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since A and B are independent, $P(A \cap B) = P(A)P(B) = 0.28$

$$\text{Thus } P(C) = 0.4 + 0.7 - 0.28 = 0.82.$$

- c. We want to obtain $P(A \cap B' | C)$

$$\begin{aligned} P(A \cap B' | C) &= \frac{P(A \cap B' \cap C)}{P(C)} = \frac{P(A \cap B' \cap (A \cup B))}{P(C)} \\ &= \frac{P(A \cap B')}{P(C)} \end{aligned}$$

Since A and B are independent, A and B' are also independent. $P(A \cap B') = P(A) \cdot P(B') = 0.4 \times 0.3 = 0.12$

$$P(A \cap B' | C) = \frac{P(A \cap B')}{P(C)} = \frac{0.12}{0.82} = 0.146$$

72. For A_1 and A_2 . $P(A_1) \cdot P(A_2) = 0.22 \times 0.25 = 0.055$
 $P(A_1 \cap A_2) = 0.11 \neq P(A_1)P(A_2)$

Thus A_1 and A_2 are dependent.

For A_1 and A_3 . $P(A_1)P(A_3) = 0.616$
 $P(A_1 \cap A_3) = 0.05 \neq P(A_1)P(A_3)$

Thus A_1 and A_3 are dependent.

For A_2 and A_3 . $P(A_2)P(A_3) = 0.07$
 $P(A_2 \cap A_3) = 0.07 = P(A_2)P(A_3)$

Thus A_2 and A_3 are independent.

$$(A_1 \cap A_2 \cap A_4) \cup (A_2 \cap A_3 \cap A_4)$$

80. Let A_i denote the event that component i works.

$$P(\text{system works}) = P((A_1 \cup A_2) \cup (A_3 \cap A_4))$$

$$= P(A_1 \cup A_2) + P(A_3 \cap A_4) - P((A_1 \cup A_2) \cap A_3 \cap A_4)$$

$$= P(A_1) + P(A_2) - P(A_1 \cap A_2) + P(A_3 \cap A_4) \\ - P(A_1 \cap A_3 \cap A_4) - P(A_2 \cap A_3 \cap A_4) \\ + P(A_1 \cap A_2 \cap A_3 \cap A_4)$$

Since they are mutually independent,

$$P(\text{system works}) = 0.9 + 0.9 - 0.9^2 + 0.9^2 - 0.9^3 - 0.9^3 + 0.9^4 \\ = 0.9981$$

84.

We can know the probability of one vehicle pass $p = 0.7$. Let corresponding letters denote corresponding event.

a. $P(A) = p^3 = 0.7^3 = 0.343$.

b. We can know A and B are mutually exclusive.

$$P(B) = 1 - P(A) = 0.657.$$

c. There exists three situations whose probabilities are equal.

$$\text{Thus } P(C) = p(1-p)^2 \cdot 3 = 0.189$$

d. The event includes one of the next three vehicles inspected passes and no vehicles pass. We use addition rules.

$$P(D) = p(1-p)^2 \cdot 3 + (1-p)^3 = 0.216$$

e. Let A_i denote i -th vehicle passes.

We want to obtain

$$\begin{aligned} & P(A_1 \cap A_2 \cap A_3 \mid A_1 \cup A_2 \cup A_3) \\ &= \frac{P(A_1 \cap A_2 \cap A_3 \cap (A_1 \cup A_2 \cup A_3))}{P(A_1 \cup A_2 \cup A_3)} \\ &= \frac{P(A_1 \cap A_2 \cap A_3)}{P(A_1 \cup A_2 \cup A_3)} \end{aligned}$$

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_2 \cap A_3) \\ &\quad - P(A_1 \cap A_3) + P(A_1 \cap A_2 \cap A_3) \end{aligned}$$

Since they are mutually independent.

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) &= P(A_1) + P(A_2) + P(A_3) - P(A_1)P(A_2) - P(A_2)P(A_3) \\ &\quad - P(A_1)P(A_3) + P(A_1)P(A_2)P(A_3) \\ &= 3p - 3p^2 + p^3 = 0.973 \end{aligned}$$

$$\text{Thus } P(A_1 \cap A_2 \cap A_3 \mid A_1 \cup A_2 \cup A_3) = \frac{0.343}{0.973} = 0.353.$$

Section 3.1

4. In China, zip code has six digits. In the code there exist at most 5 zero digits. Thus the possible values of X are 1, 2, 3, 4, 5, 6.

Three outcomes:

$$X(511400) = 4$$

$$X(510100) = 3$$

$$X(257131) = 6$$

5. No. Suppose S is the set of integers.

$$X(s) = \begin{cases} 0, & \text{if } s \text{ is an integer} \\ 1, & \text{otherwise.} \end{cases}$$

Then the set of possible values of X is $\{0, 1\}$, which is not an infinite set.

8. $Y = 3, 4, 5, 6, 7, \dots$

For $Y=3$: SSS

For $Y=4$: FSSS

For $Y=5$: SFSSS, FFSSS

For $Y=6$: SFFSSS, FSFSSS, SSFSSS, FFFSSS

For $Y=7$: SSFFSSS, SFSFSSS, FSSFSSS, SFFFSSS,
FSFFSSS, FFSFSSS, FFFFSSS.

10.

a. Possible values of T : 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

b. Possible values of X : -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6

c. Possible values of U : 0, 1, 2, 3, 4, 5, 6

d. Possible values of Z : 0, 1, 2

Section 3.2

12. a.
$$P(Y \leq 50) = p(45) + p(46) + p(47) + p(48) + p(49) + p(50)$$
$$= 0.05 + 0.10 + 0.12 + 0.14 + 0.25 + 0.17$$
$$= 0.83$$

b.
$$P(Y > 50) = p(51) + p(52) + p(53) + p(54) + p(55)$$
$$= 0.06 + 0.05 + 0.03 + 0.02 + 0.01$$
$$= 0.17$$

c. For the first person on the standby list, the number of passengers should be less than 49.

$$P(Y \leq 49) = p(45) + p(46) + p(47) + p(48) + p(49)$$
$$= 0.05 + 0.10 + 0.12 + 0.14 + 0.25$$
$$= 0.66$$

For the third person,

$$P(Y \leq 47) = p(45) + p(46) + p(47)$$
$$= 0.05 + 0.10 + 0.12$$
$$= 0.27$$

23. a.
$$P(X=2) = F(2 \leq x < 3) - F(1 \leq x < 2)$$
$$= 0.39 - 0.19$$
$$= 0.20$$

b.
$$P(X > 3) = 1 - F(3 \leq x < 4)$$
$$= 1 - 0.67$$
$$= 0.33$$

c.
$$P(2 \leq X \leq 5) = F(5 \leq x < 6) - F(1 \leq x < 2)$$
$$= 0.97 - 0.19$$
$$= 0.78$$

d.
$$P(2 < X < 5) = F(4 \leq x < 5) - F(2 \leq x < 3)$$
$$= 0.92 - 0.39$$
$$= 0.53$$

$$25. y=0: p(0) = P(B) = p$$

$$y=1: p(1) = P(GB) = (1-p) \cdot p.$$

$$\vdots$$

$$\text{Thus, } p(y) = P(GG \cdots GB) = (1-p)^y \cdot p \text{ for } y=0,1,2,\dots$$

